

Geometry
Unit 13
Lesson 3 Homework

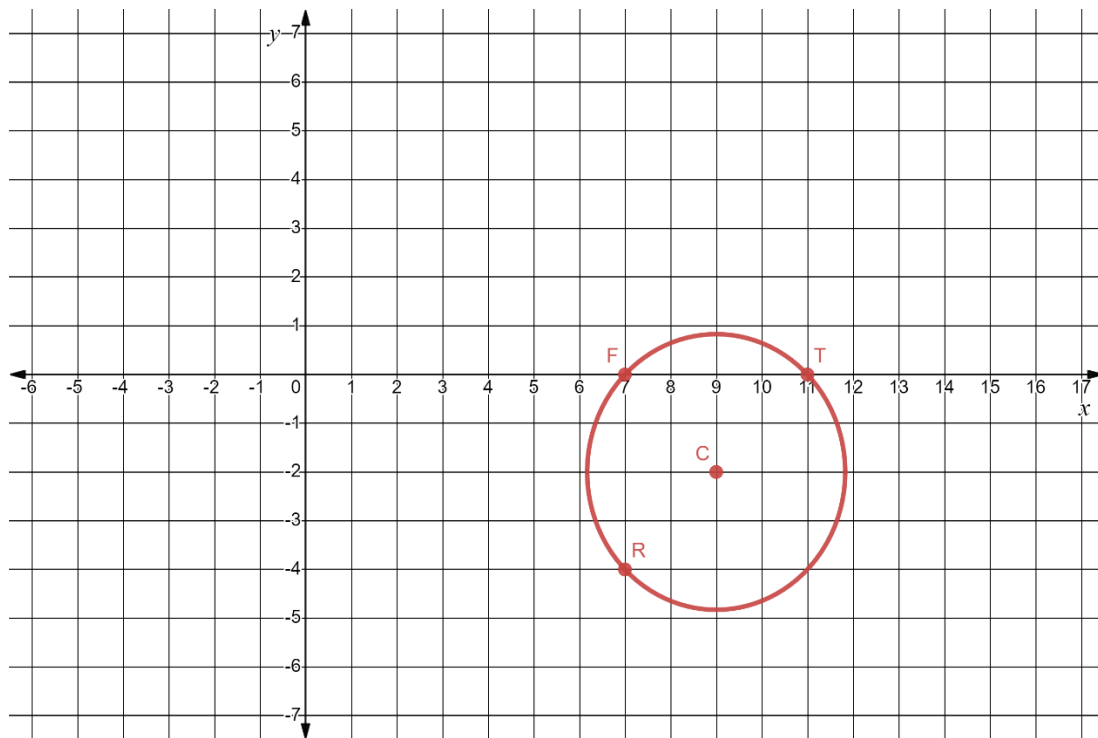
Name Answer Key
Date _____
Period _____

Use the following information for questions 1-4. The equation of Circle C is $(x - 9)^2 + (y + 2)^2 = 8$.

1. Complete the table by substituting the given coordinates for x and y in the equation of Circle C . Then, determine the value of the left side of the equation to complete the simplified equation in the last column.

Coordinate	Substitution	Simplified Equation
$F(7,0)$	$(7 - 9)^2 + (0 + 2)^2 = (-2)^2 + (2)^2$ $= 4 + 4 = 8$	$8 = 8$
$R(7, -4)$	$(7 - 9)^2 + (-4 + 2)^2 = (-2)^2 + (-2)^2$ $= 4 + 4 = 8$	$8 = 8$
$T(11,0)$	$(11 - 9)^2 + (0 + 2)^2 = (2)^2 + (2)^2$ $= 4 + 4 = 8$	$8 = 8$

2. Graph Circle C , $(x - 9)^2 + (y + 2)^2 = 8$.



3. Plot $F(7,0)$, $R(7,-4)$, $T(11,0)$ on the circle.
4. Determine the length of the diameter of Circle C .

$$d = 2 \cdot \sqrt{8} = 4\sqrt{2} \approx 5.66$$

Use the following information for questions 5-7. Circle P has a center located at $T(3,4)$ and point $L(4,9)$ on the circle.

5. Determine two points that are the endpoints of a diameter of Circle P .

Answers may vary. $(3,4) = \left(\frac{4+x}{2}, \frac{9+y}{2}\right) \rightarrow \begin{array}{l} 4+x=6 \\ x=2 \end{array} \quad \begin{array}{l} 9+y=8 \\ y=-1 \end{array}$
 $(4,9)$ and $(2,-1)$

6. What is the length of the radius?

$T(3,4)$ and $L(4,9)$
 $\sqrt{(3-4)^2 + (4-9)^2} = \sqrt{26}$

7. Write the equation for Circle P .

$(x-3)^2 + (y-4)^2 = 26$

The equation of Circle W is $(x-3)^2 + (y+5)^2 = 17$. Determine whether the following points lie inside, outside, or on Circle W .

Point	Work	Location
$C(3, -10)$	$(3-3)^2 + (-10+5)^2 = 25 > 17$	☐ Inside Circle W ☒ Outside Circle W ☐ On Circle W
$N(2,3)$	$(2-3)^2 + (3+5)^2 = 65 > 17$	☐ Inside Circle W ☒ Outside Circle W ☐ On Circle W
$T(4,-1)$	$(4-3)^2 + (-1+5)^2 = 17 = 17$	☐ Inside Circle W ☐ Outside Circle W ☒ On Circle W

8. The equation of Circle R is $(x+3)^2 + (y+6)^2 = 10$.

Select all points that are outside Circle R .

- ☐ $(-3,-6)$ $(-3+3)^2 + (-6+6)^2 = 0 < 10$ inside
☒ $(-3,-2)$ $(-3+3)^2 + (-2+6)^2 = 16 > 10$ outside
☐ $(0,-5)$ $(0+3)^2 + (-5+6)^2 = 10 = 10$ on
☒ $(0,0)$ $(0+3)^2 + (0+6)^2 = 45 > 10$ outside
☐ $(-1,-6)$ $(-1+3)^2 + (-6+6)^2 = 4 < 10$ inside

9. Circle M and Circle T intersect at point H .

Equation of Circle M is $x^2 + (y-2)^2 = 1$

Equation of Circle T is $(x+3)^2 + (y-1)^2 = 9$

Which coordinate pair is the location of point H ?

- ☐ $(-3,4)$ $(-3)^2 + (4-2)^2 = 13 > 1 \rightarrow$ not on circle M
☐ $(4,0)$ $(4)^2 + (0-2)^2 = 20 > 1 \rightarrow$ not on circle M
☐ $(0,3)$ $(0)^2 + (3-2)^2 = 1 \checkmark$, $(0+3)^2 + (3-1)^2 = 13 > 9 \rightarrow$ not on circle T
☒ $(0,1)$ $(0)^2 + (1-2)^2 = 1 = 1 \checkmark$, $(0+3)^2 + (1-1)^2 = 9 = 9 \checkmark$